Stormwater Regulation Summary: Boston, MA

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary drivers: Combined sewer overflow (CSO) reduction, Charles River and Boston Harbor water quality.
- Regulated by BWSC (Boston Water & Sewer Commission) for site drainage + MassDEP Stormwater Standards (10 standards).
- Article 37 (Green Building) applies to projects >50,000 SF GFA — LEED certifiable standard required.
- *New CSS connections may require I/I (Inflow/Infiltration) reduction ratio (verify specific ratio with BWSC) — makes stormwater management mandatory.*
- Net Zero Carbon Zoning (effective July 2025) requires zero-emission new construction — multiple compliance pathways including solar, renewable energy contracts, and low-carbon design.

2. Typical Soil Conditions

- Boston sits on a mix of glacial till, marine clay (Boston Blue Clay), and historic fill (Back Bay, Seaport).
- Boston Blue Clay is a well-known marine deposit — extremely low permeability, HSG D.
- Hydrologic Soil Group: Predominantly C and D; some sandy outwash areas (B) in outer neighborhoods.
- Back Bay, Seaport, and East Boston are largely built on fill over tidal flats — variable and often poor infiltration.
- Implication: Ground infiltration is rarely viable — retention via rooftop systems and controlled release to sewers is the practical approach.
- *Groundwater levels are high in waterfront areas — subsurface BMPs face additional constraints.*

3. Green Roof Requirements

- Green roofs are not explicitly mandatory, but Article 37 (LEED certifiable) and BERDO (Building Emissions Reporting) create strong incentive.
- BWSC provides up to 30% stormwater fee credit for green roofs that manage on-site runoff.
- Net Zero Carbon Zoning (2025) requires zero-emission new construction — bio-solar (green roof + PV) is a strong compliance pathway.
- BERDO requires buildings ≥20,000 SF to meet emissions reduction targets — green roofs contribute to compliance.
- MassDEP Stormwater Standard #4 (80% TSS removal) can be partially met with green roof systems.
- MA SMART 3.0 solar incentive provides 10-20 year performance payments — stacks with green roof stormwater credits.

4. TSS Requirements

- MassDEP Stormwater Standard #4 requires 80% TSS removal (new developments may require 90% under updated standards).
- Green roofs are recognized as a BMP contributing to TSS removal — typically credited at 80%+ reduction for captured volume.
- Standard #4 also requires 44% phosphorus removal — green roofs provide nutrient removal credit.
- For projects in the Charles River watershed, additional phosphorus loading limits may apply.
- BWSC may require additional treatment beyond green roof for high-TSS source areas (parking, loading).
- *MassDEP BMP crediting varies by system design — verify current credit rates in the MA Stormwater Handbook.*

5. Nature-Based Retention Requirements

- Retain 1.0" for sites <100,000 SF; retain 1.25" for sites ≥100,000 SF.
- Retention volume must be managed through on-site BMPs prior to discharge to BWSC system.
- CN (Curve Number) method per TR-55 is the standard design approach.
- MassDEP Stormwater Standard #3: No new stormwater conveyances to wetlands or waterbodies without treatment.
- Standard #2: Post-development peak discharge ≤ pre-development for 2-yr and 10-yr storms (10-yr = 5.15" in Boston).
- *I/I reduction ratio for new CSS connections — verify specific ratio requirements with BWSC for the project sewer shed.*

6. Allowable Outflow Rates

- 2-year storm: Post-development peak ≤ pre-development peak discharge.
- 10-year storm (5.15"): Post-development peak ≤ pre-development peak discharge.
- BWSC does not use a flat l/s/ha rate — requirements are relative to pre-development conditions.
- For CSS areas, the 4:1 I/I ratio effectively limits new discharge to 25% of pre-existing inflow contribution.
- *Typical allowable rates vary significantly by sewer shed — BWSC approval required for each connection.*
- *For reference, typical pre-development rates for urban Boston sites: 20-50 l/s/ha range for the 10-yr storm.*